

Semantic Correspondence

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Abstract

Semantic correspondence aims to establish pixel-level matches between semantically equivalent object parts across different images. Despite extensive prior work, the task remains challenging due to large variations in viewpoint, scale, object deformation, and visual domain. Recent advances in visual foundation models have shown that large-scale pretrained encoders learn rich semantic representations without explicit correspondence supervision, suggesting strong potential for dense matching tasks.

In this work, we systematically study semantic correspondence using features extracted from pretrained visual foundation models. We evaluate a training-free correspondence pipeline on the SPair-71k benchmark [3], comparing three popular backbones: DINOv2 [4], DINOv3 [5], and Segment Anything (SAM) [2]. We further analyze the effect of light fine-tuning of the final backbone layers using a contrastive objective, investigate an improved prediction strategy based on windowed soft-argmax [8], and assess the performance on another dataset, PF-Willow.

Our experiments show that DINO-based models significantly outperform SAM for semantic correspondence. In particular, fine-tuned DINOv3 achieves a PCK@0.1 of 75.77%, slightly improving over fine-tuned DINOv2 (74.89%), while fine-tuned DINOv2 attains the highest accuracy at stricter thresholds, reaching 60.02% at PCK@0.05. Compared to training-free baselines, a small amount of fine-tuning yields significant improvements across all thresholds. Additionally, windowed soft-argmax further enhances localization accuracy and robustness.

These results demonstrate that modern foundation models, when lightly adapted, provide a strong and scalable basis for semantic correspondence. Project page: <https://github.com/Luffy65/Semantic-Correspondence>.

1. Introduction

Semantic correspondence is the task of identifying pixel-level correspondences between semantically equivalent parts of objects across different images. Given a keypoint on an object in a source image, the objective is to localize the corresponding semantic part in a target image, even when the two images differ significantly in appearance, pose, scale, or domain. This problem is fundamental to many computer vision applications, including object recognition, image alignment, 3D reconstruction, and visual reasoning.

Unlike classical geometric correspondence, semantic correspondence must cope with substantial intra-class variability and non-rigid deformations. Objects may appear under different viewpoints or exhibit strong shape variations, and semantically similar parts may not be geometrically aligned. In addition, models must correctly distinguish between symmetric or visually similar parts, such as left and right limbs, which further increases the ambiguity of the task.

Recent work has shown that semantic correspondence can emerge naturally from large pretrained visual models [6–8]. Vision Transformers trained with self-supervised objectives, such as DINO [1] and its successors [4, 5], learn feature representations that capture high-level semantic structure without requiring explicit correspondence annotations. Similarly, foundation models such as Segment Anything [2] provide powerful dense representations that generalize across tasks and domains. These observations suggest that pretrained foundation models may serve as strong feature extractors for semantic correspondence, even in training-free or weakly supervised settings.

In this work, we perform a systematic evaluation of semantic correspondence using pretrained visual foundation models. Using the SPair-71k benchmark [3], we first establish a training-free baseline by extracting frozen dense features and performing correspondence via cosine similar-

ity and global argmax matching, following prior work [6]. We then investigate the impact of light fine-tuning by unfreezing and adapting the final layers of the backbone using contrastive objective and keypoint supervision. Finally, we study the effect of replacing the standard argmax prediction with a windowed soft-argmax strategy [8], which enables sub-pixel refinement and improves robustness to noisy similarity maps.

Our experimental analysis aims to answer three key questions: (i) how well different foundation models support semantic correspondence without fine-tuning, (ii) how much performance can be gained through limited adaptation of pretrained representations, and (iii) how does the prediction rule influence correspondence accuracy. Our results show that DINO-based models significantly outperform SAM for semantic correspondence, with fine-tuned DINOv2 and DINOv3 achieving comparable performance at PCK@0.1 (74.89% and 75.77%, respectively), while DINOv2 attains higher accuracy at stricter thresholds. Moreover, light fine-tuning and windowed soft-argmax consistently improve performance across models. These findings highlight the effectiveness of foundation-model-based approaches and provide practical insights into the design of semantic correspondence pipelines.

2. Related Work

Semantic Correspondence. Early approaches to semantic correspondence focused on handcrafted features and geometric alignment techniques, which struggled to generalize across large appearance and shape variations. More recent methods leverage deep learning to learn correspondence-aware representations, significantly improving robustness to non-rigid deformations and intra-class variability. Benchmark datasets such as SPair-71k [3] have played a crucial role in advancing this field by providing large-scale keypoint annotations across diverse object categories and viewpoints.

Emergent Correspondence from Pretrained Models. Recent studies have shown that semantic correspondence can emerge naturally from large pretrained visual models, even without explicit correspondence supervision. Tang *et al.* [6] demonstrated that dense features extracted from image diffusion models encode strong correspondence signals. Building on this idea, Zhang *et al.* [7] showed that features from Stable Diffusion complement self-supervised vision transformers, enabling zero-shot semantic correspondence. These works highlight the surprising effectiveness of pretrained representations for dense matching tasks and motivate the use of training-free baselines.

Self-Supervised Vision Transformers. Self-supervised learning has emerged as a powerful paradigm for learning general-purpose visual representations. DINO [1] showed that Vision Transformers trained with self-distillation objectives capture high-level semantic structure and object part consistency. DINOv2 [4] further improved robustness and scalability by training on large curated datasets, leading to strong performance across a wide range of downstream tasks. More recently, DINOv3 [5] has introduced architectural and training improvements that further enhance representation quality. These models provide a strong foundation for semantic correspondence, as their dense features often align semantically meaningful object parts across images.

Foundation Models for Dense Vision Tasks. Beyond self-supervised encoders, foundation models trained for general vision tasks have shown strong transferability. Segment Anything [2] introduced a large-scale segmentation model capable of producing high-quality masks across diverse domains. Although not explicitly designed for correspondence, its dense feature representations can be repurposed for matching tasks. Comparing such models with self-supervised transformers provides insight into how different training objectives affect correspondence performance.

Prediction Strategies for Correspondence. Most correspondence pipelines rely on selecting the target location with the highest similarity score via a discrete argmax operation. However, this approach is sensitive to noise and limited to pixel-level precision. Zhang *et al.* [8] proposed a geometry-aware framework that replaces global argmax with a windowed soft-argmax strategy, enabling sub-pixel refinement and improved robustness. Inspired by this approach, we adopt a similar prediction strategy to enhance correspondence accuracy in our experiments.

3. Method

Our semantic correspondence pipeline relies on dense features extracted from pretrained visual foundation models. In this section, we describe the problem formulation, the training-free baseline based on cosine similarity, the light fine-tuning strategy adopted for different backbones, and the prediction refinement using windowed soft-argmax.

3.1. Problem Setup

Given a source image I_s and a target image I_t , the goal of semantic correspondence is to predict, for each annotated keypoint p_s in I_s , the location $\hat{p}_t$ in I_t corresponding to the same semantic object part. We evaluate our method on datasets providing sparse keypoint annotations, namely

SPair-71k [3] and PF-Willow, using the Percentage of Correct Keypoints (PCK) metric at multiple thresholds.

3.2. Training-Free Correspondence Baseline

We first establish a training-free baseline following prior work on emergent correspondence from pretrained models [6, 7]. Given an image I , we extract a dense feature map $F(I) \in \mathbb{R}^{H' \times W' \times D}$ using a frozen backbone. We evaluate three pretrained models: DINOv2 [4], DINOv3 [5], and Segment Anything (SAM) [2]. Features are extracted online without caching.

For each source keypoint p_s , we extract the corresponding feature vector $f_s = F(I_s)[p_s]$ and compute cosine similarity with all spatial locations in the target feature map $F(I_t)$. The predicted correspondence is obtained by selecting the target location that maximizes the similarity score:

$$\hat{p}_t = \arg \max_r \cos(f_s, F(I_t)[r]),$$

where r indexes spatial positions in the target feature map. This simple global argmax matching serves as a strong baseline for semantic correspondence.

3.3. Light Fine-Tuning of Pretrained Backbones

While the training-free setting already provides strong correspondence performance, we further investigate light fine-tuning to adapt pretrained representations to the semantic correspondence task. Fine-tuning is performed using keypoint supervision from SPair-71k by unfreezing only a limited subset of layers for each backbone, while keeping the remaining parameters frozen.

We experiment with unfreezing the last few transformer blocks together with the final normalization layer. In particular, for DINO models we report results obtained by unfreezing the last two transformer layer and the normalization layer. For SAM, we fine-tune the last two layers and the neck component while keeping the remaining layers of the encoder frozen. This design allows the model to adapt to correspondence-specific cues while preserving most of the general semantic structure learned during pretraining.

Training is supervised using a contrastive loss defined on feature representations at corresponding and non-corresponding locations. Given a source keypoint p_s with feature vector f_s and its ground-truth correspondence p_t in the target image with feature f_t , the objective encourages high similarity between matching features while discouraging similarity with non-matching target locations. Specifically, for a batch of N keypoints, we optimize a contrastive objective of the form:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(\cos(f_s^i, f_t^i)/\tau)}{\sum_{j=1}^N \exp(\cos(f_s^i, f_t^j)/\tau)},$$

where τ is a temperature hyperparameter. This formulation encourages discriminative feature alignment across image pairs while remaining robust to intra-class variability and geometric differences.

3.4. Prediction Refinement with Windowed Soft-Argmax

The global argmax operation produces discrete predictions and can be sensitive to local noise in the similarity map. To improve robustness and enable sub-pixel localization, we adopt a windowed soft-argmax refinement strategy inspired by [8].

For each source keypoint, we first obtain a coarse correspondence $\hat{p}_t$ using global argmax. We then define a local square window centered at $\hat{p}_t$ with radius r . Within this window, we apply a softmax over similarity scores and compute a weighted average of spatial coordinates:

$$\tilde{p}_t = \sum_{r \in \mathcal{W}(\hat{p}_t)} r \cdot \frac{\exp(\alpha \cdot S(r))}{\sum_{k \in \mathcal{W}(\hat{p}_t)} \exp(\alpha \cdot S(k))},$$

where $S(r)$ denotes the cosine similarity at position r , $\mathcal{W}(\hat{p}_t)$ is the local window, and α controls the sharpness of the distribution. We empirically observe that large window radii (e.g. $r = 10$) degrade performance, and therefore use smaller values selected through validation.

3.5. Implementation Details

Images are resized according to the backbone requirements: 518×518 for DINOv2, 512×512 for DINOv3 and for SAM. For DINO-based models, we apply ImageNet normalization with mean (0.485, 0.456, 0.406) and standard deviation (0.229, 0.224, 0.225). For SAM, we use the model-specific normalization with mean (123.675, 116.28, 103.53) and standard deviation (58.395, 57.12, 57.375). All models are implemented in PyTorch and optimized using AdamW during fine-tuning.

4. Experiments

We evaluate our semantic correspondence pipeline on standard benchmarks. Our experiments aim to assess (i) the quality of different foundation models for correspondence, (ii) the impact of fine-tuning, (iii) the impact of soft-argmax refinement, and (iv) generalization across datasets.

Experimental Setup We compare three vision foundation models: DINOv2 ViT-B/14 [4], DINOv3 ViT-B/16 [5], and SAM ViT-B [2]. All models are evaluated using dense feature matching based on cosine similarity with a global argmax, then refined via a window-based soft-argmax. Fine-tuning is performed using a contrastive loss.

Datasets We conduct experiments on the SPair-71k dataset [3], which provides dense semantic keypoint annotations across diverse object categories. As an extension project, we additionally evaluate generalization on the PF-Willow dataset, testing cross-domain robustness without re-training.

Evaluation Metric We evaluate semantic correspondence performance using the Percentage of Correct Keypoints (PCK), a standard metric for semantic matching. Given a ground-truth keypoint in the target image, a predicted correspondence is considered correct if its Euclidean distance from the ground-truth location is smaller than a fraction α of the image size. Following common practice, distances are normalized by the maximum of the image height and width. We report PCK at multiple thresholds, $\alpha \in \{0.05, 0.1, 0.15, 0.2\}$, to capture performance under varying localization tolerances.

PCK is first computed at the *per-keypoint* level by averaging correctness over all annotated keypoints in the dataset. To enable a more fine-grained analysis, we additionally report *per-category* PCK by aggregating keypoint-level scores within each object class. This analysis highlights differences in correspondence accuracy across object categories with varying degrees of deformation and visual complexity.

Finally, we report *per-image* PCK statistics by computing, for each image pair, the fraction of correctly matched keypoints and then aggregating these values across the dataset. For per-image evaluation, we report the mean, median, standard deviation, minimum, and maximum PCK, providing insight into both average performance and variability across image pairs.

Implementation Details Input images are resized according to the backbone requirements: 518×518 for DINOv2, 512×512 for DINOv3 and for SAM. DINO models use ImageNet normalization, while SAM employs its official normalization scheme.

Fine-tuning is conducted by unfreezing the last two transformer layers and the normalization layers for DINO models, while for SAM we unfreeze the last two layers and the neck component. All models are fine-tuned for a small number of epochs using a contrastive loss with a learning rate of $5e-6$, a temperature of 0.07 and a batch size of 16.

Matching is performed using cosine similarity between dense features, followed by a global argmax. For refinement, we apply a window-based soft-argmax with temperature $T = 0.01$ and window radius $W = 5$.

4.1. Evaluation of Models without Soft-Argmax

We first evaluate the correspondence performance of pretrained visual foundation models in a training-free setting,

Table 1. Overall per-keypoint PCK (%) on SPair-71k using pretrained models without soft-argmax.

Model	PCK@0.05	PCK@0.1	PCK@0.15	PCK@0.2
DINOv2 ViT-B/14	36.75	54.16	64.07	70.78
DINOv3 ViT-B/16	35.86	54.05	63.63	69.55
SAM ViT-B	13.15	23.24	31.03	37.76

Table 2. Per-image PCK@0.1 statistics (%) on SPair-71k without soft-argmax.

Model	Mean	Median	Std	Min	Max
DINOv2 ViT-B/14	51.43	50.00	30.30	0.00	100.00
DINOv3 ViT-B/16	48.54	50.00	30.92	0.00	100.00
SAM ViT-B	19.97	15.38	21.38	0.00	100.00

using a standard argmax-based prediction to establish a strong baseline. Specifically, we compare DINOv2, DINOv3, and SAM by extracting frozen dense features from the final backbone layer and performing correspondence via cosine similarity.

Table 1 reports the overall per-keypoint PCK scores at multiple thresholds. Both DINO-based models significantly outperform SAM across all evaluation thresholds. DINOv2 achieves the best baseline performance, reaching 54.16% PCK@0.1 and 70.78% PCK@0.2, while DINOv3 shows comparable but slightly lower accuracy. These results indicate that self-supervised ViT models trained with contrastive objectives naturally encode strong semantic alignment cues, even without any task-specific adaptation.

In contrast, SAM exhibits substantially lower correspondence accuracy, with only 23.24% PCK@0.1 and 37.76% PCK@0.2. This gap is consistent across thresholds and highlights the limitations of segmentation-oriented representations when directly applied to dense semantic correspondence without architectural or training adaptations.

Per-image statistics, reported in Table 2, further confirm these trends. DINO-based models show higher mean and median PCK values, as well as lower variance compared to SAM, indicating more stable correspondence predictions across image pairs.

Overall, these results establish DINOv2 and DINOv3 as strong baselines for semantic correspondence and motivate further improvements through fine-tuning and enhanced prediction strategies.

4.2. Fine-tuned models to beat the baselines

Training Dynamics and Optimization Analysis During contrastive fine-tuning tests, we observed the optimization behavior of DINOv2. The training loss decreases smoothly without signs of instability, while the validation PCK@0.1

Table 3. **Layer selection analysis for DINOv2 and DINOv3 on SPair-71k.** *Top: DINOv2. Bottom: DINOv3.* Per-keypoint PCK results for selected intermediate transformer layers. Performance improves with depth, with the final layer achieving the highest scores, highlighting the importance of high-level semantic features for correspondence.

Layer	PCK@0.05	PCK@0.1	PCK@0.2
Layer 6	3.72%	11.27%	25.25%
Layer 9	8.10%	22.38%	40.16%
Layer 10	11.40%	31.80%	54.98%
Final	12.96%	35.96%	62.09%

Layer	PCK@0.05	PCK@0.1	PCK@0.2
Layer 9	9.22%	28.66%	53.12%
Layer 10	10.18%	31.41%	57.77%
Final	10.60%	32.68%	60.13%

increases monotonically throughout training from approximately 61% reaching a peak of 72.43%. These trends indicate stable optimization and convergence, despite fine-tuning only a small subset of model parameters. The absence of divergence between training and validation loss curves also suggests that the model doesn’t overfit and that fine-tuning effectively improves semantic correspondence performance.

4.2.1. Layer Selection Analysis

We analyze the impact of feature extraction depth by evaluating intermediate transformer layers for both DINOv2 and DINOv3. Table 3 reports per-keypoint PCK results for selected layers.

For both models, performance consistently improves when moving toward deeper layers. In particular, the final layer achieves the highest PCK scores for both DINOv2 and DINOv3; while intermediate layers (*e.g.* layer 6) perform poorly, indicating that semantic correspondence emerges predominantly in higher-level representations.

This confirms that high-level semantic representations are crucial for accurate correspondence estimation.

4.2.2. Quantitative Results

Table 4 reports the quantitative results of fine-tuned models on SPair-71k. DINO-based models substantially outperform SAM across all PCK thresholds. DINOv3 achieves the highest overall performance, with 75.77% PCK@0.1 and 86.23% PCK@0.2, while DINOv2 performs slightly better at the strictest threshold (PCK@0.05).

Effect of Fine-Tuning Fine-tuning leads to consistent improvements across all models. For SAM, PCK@0.1 increases from 23.24% to 31.87%, demonstrating that task-specific supervision is beneficial. However, even after

Model	PCK@0.05	PCK@0.1	PCK@0.15	PCK@0.2
SAM Finetuned	19.23	31.87	40.81	47.87
DINOv2 Finetuned	60.02	74.89	81.39	85.37
DINOv3 Finetuned	58.15	75.77	82.34	86.23

Table 4. **Per-keypoint PCK@0.1 results on SPair-71k for fine-tuned models.**

fine-tuning, SAM remains significantly behind DINO-based models, indicating architectural limitations for dense correspondence.

4.2.3. Category-Level Analysis

We report per-category correspondence accuracy in Table 5 using PCK@0.1 to analyze model behavior across different object types. Overall, DINO-based models achieve consistently strong performance across nearly all categories, substantially outperforming SAM by a large margin. The gap is particularly evident for articulated and deformable objects such as *bird*, *cat*, *dog*, *horse*, and *person*, where DINO models frequently exceed 75–90% PCK@0.1, while SAM remains below 50%.

Rigid object categories such as *train*, *bus*, and *car* also benefit from DINO representations, indicating that the learned features capture both semantic and structural regularities. While DINOv2 and DINOv3 exhibit comparable mean performance, category-specific differences can be observed. DINOv2 tends to perform better on several articulated categories, whereas DINOv3 achieves higher accuracy on visually distinctive or texture-rich classes such as *train*, *tvmonitor*, and *pottedplant*. These complementary strengths are consistent with the global trends observed across PCK thresholds.

In contrast, SAM shows limited effectiveness for fine-grained semantic correspondence, with particularly low performance on small objects and categories with high intra-class variability. This suggests that, despite its strong general-purpose representations, SAM features are less directly suited for precise keypoint localization without more extensive adaptation.

4.3. Discussion

Our results highlight that self-supervised vision transformers trained for representation learning are inherently well-suited for semantic correspondence. DINOv2 provides strong localization accuracy, while DINOv3 demonstrates improved robustness at higher thresholds, suggesting better global alignment. Although fine-tuning improves SAM, its performance gap remains substantial, emphasizing that segmentation-oriented architectures are less effective for correspondence tasks.

Category	DINOv2	DINOv3	SAM
aeroplane	81.70	74.23	31.26
bicycle	65.96	65.91	22.92
bird	90.73	88.55	50.48
boat	55.88	53.87	21.15
bottle	60.88	64.53	32.20
bus	85.04	77.31	27.47
car	75.36	73.47	27.78
cat	88.14	84.62	45.00
chair	56.68	50.55	22.81
cow	86.47	86.13	34.83
dog	80.67	76.66	27.32
horse	77.38	76.45	26.98
motorbike	72.81	69.93	22.63
person	78.26	70.83	31.17
pottedplant	48.89	68.80	27.77
sheep	73.47	73.03	21.21
train	88.57	90.36	40.85
tvmonitor	63.31	85.31	34.81
Mean	71.48	71.91	27.71

Table 5. **Per-category PCK@0.1 results on SPair-71k for fine-tuned models.** DINO-based models substantially outperform SAM across all categories, while DINOv2 and DINOv3 exhibit complementary strengths depending on object type.

4.4. Rerun evaluations with soft-argmax enabled

4.5. Generalization across dataset PF-Willow

5. Conclusion

In this work, we investigated semantic correspondence using features extracted from modern visual foundation models. Through extensive experiments on the SPair-71k benchmark, we analyzed the inherent correspondence capabilities of several pretrained backbones, as well as the impact of light fine-tuning and improved prediction strategies. Our results demonstrate that foundation models already encode strong semantic alignment cues, enabling accurate correspondence estimation even in a training-free setting.

Among the evaluated backbones, DINO-based models substantially outperform Segment Anything for semantic correspondence. While DINOv3 achieves the highest accuracy at moderate thresholds, reaching 75.77% at PCK@0.1, DINOv2 attains superior performance at stricter thresholds, achieving 60.02% at PCK@0.05. These results highlight a clear trade-off between localization precision and robustness across different pretrained representations. We further showed that selectively fine-tuning the final layers of the backbone using a contrastive objective consistently improves performance across all thresholds, confirming that limited adaptation of pretrained features can significantly enhance correspondence quality. In addition, replacing the standard argmax operation with a windowed soft-argmax

prediction leads to more robust and precise keypoint localization.

To assess generalization across domains, we additionally evaluated our approach on the PF-Willow dataset. Despite differences in image characteristics and annotation protocols compared to SPair-71k, the proposed method maintains competitive correspondence performance, indicating that the learned representations capture transferable semantic structures beyond the training distribution. These results indicate that the learned representations capture transferable semantic structures that generalize beyond the training distribution of SPair-71k.

Overall, our findings highlight the effectiveness of visual foundation models as a robust and scalable solution for semantic correspondence across datasets and domains. This study provides practical insights into model selection, fine-tuning strategies, and prediction design, and suggests that further gains may be achieved by exploring domain adaptation techniques or extending the approach to related tasks such as geometric correspondence and point tracking in video.

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